Higher-Order Differential Equations

A. Haghighat M.

Department of Building, Civil and Environmental Engineering

ENGR 213

Lecture 12 (October 23, 2026)

Today's Lecture

- Undetermined Coefficients
  - Method of Undetermined Coefficients

Reference

Advanced Engineering Mathematics, D.G. Zill (7th edition), **Section 3.4**

- **Nonhomogeneous linear DEs**

  - To solve a nonhomogeneous linear DE with constant coefficients:

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = g(x)$$

where a_i, i = 0, 1, …, n are real constants and $a_n \neq 0$.

we must:

I. Find the complementary function $\mathbf{y_c}$ of the associated homogeneous DE

$$(a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = 0)$$

II. Find any particular solution $\mathbf{y_p}$ of the nonhomogeneous DE

III. Construct the general solution $\mathbf{y = y_c + y_p}$

- **Method of Undetermined Coefficients**

  - This method can be used to obtain a particular solution $\mathbf{y_p}$.
  - The underlying idea in this method is to make an educated guess about the form of $\mathbf{y_p}$ based on the kinds of functions that make up g(x).
  - The method is limited to nonhomogenous linear DEs where
    - the coefficients, a_i, i = 0, 1,…, n are constants
    - g(x) is a constant, a polynomial function, exponential function ($e^{\alpha x}$), sine or cosine functions (sin βx or cos βx), or finite sums and products of these functions.


QUESTIONS

EXAMPLE 1

Solve $y'' + 4y' - 2y = 2x^2 - 3x + 6$

EXAMPLE 2

Find a particular solution of $y'' - y' + y = 2 \sin 3x$

EXAMPLE 3

Solve $y'' - 2y' - 3y = 4x - 5 + 6xe^{2x}$

EXAMPLE 4

Find a particular solution of $y'' - 5y' + 4y = 8e^x$


QUESTIONS

Table below illustrates some specific examples of $g(x)$ along with the corresponding form of the particular solution

TABLE 3.4.1. Trial Particular Solutions

$g(x)$	Form of y_p
1. 1 (any constant)	A
2. $5x + 7$	$Ax + B$
3. $3x^2 - 2$	$Ax^2 + Bx + C$
4. $x^3 - x + 1$	$Ax^3 + Bx^2 + Cx + E$
5. $\sin 4x$	$A \cos 4x + B \sin 4x$
6. $\cos 4x$	$A \cos 4x + B \sin 4x$
7. e^{5x}	Ae^{5x}
8. $(9x - 2)e^{5x}$	$(Ax + B)e^{5x}$
9. x^2e^{5x}	$(Ax^2 + Bx + C)e^{5x}$
10. $e^{3x} \sin 4x$	$Ae^{3x} \cos 4x + Be^{3x} \sin 4x$
11. $5x^2 \sin 4x$	$(Ax^2 + Bx + C) \cos 4x + (Ex^2 + Fx + G) \sin 4x$
12. $xe^{3x} \cos 4x$	$(Ax + B)e^{3x} \cos 4x + (Cx + E)e^{3x} \sin 4x$

Case I: No function in the assumed particular solution is a solution of the associated homogeneous differential equation.

EXAMPLE 5

Determine the from of a particular solution of $y'' - 8y' + 25y = 5x^3 e^{-x} - 7e^{-x}$

EXAMPLE 6

Determine the from of a particular solution of $y'' + 4y = x \cos x$

Form Rule for Case I: If g(x) consists of a sum of *m* terms (of the kind listed in the table), then the assumption for a particular solution y_p consists of the sum of the trial forms $y_{p_1}, y_{p_2}, \ldots, y_{p_m}$ corresponding to these terms:

$$\boldsymbol{y_p = y_{p_1} + y_{p_2} + \cdots + y_{p_m}}$$

EXAMPLE 7

Determine the from of a particular solution of

$y'' - 9y' + 14y = 3x^2 - 5 \sin 2x + 8xe^{6x}$

Case II: A function in the assumed particular solution is also a solution of the associated homogeneous differential equation.

EXAMPLE 8

Find a particular solution of $y'' - 2y' + y = e^x$

Multiplication Rule for Case II: If any y_{p_i} contains terms that duplicate terms in y_c, then that y_{p_i} must be multiplied by x^n, where n is the smallest positive integer that eliminates that duplication.


QUESTIONS

EXAMPLE 9

Determine the form of a particular solution of $y'' - 6y' + 9y = 6x^2 + 2 - 12e^{3x}$

EXAMPLE 10

Determine the form of a particular solution of $y^{(4)} + y''' = 1 - x^2e^{-x}$

EXAMPLE 11

Solve the initial-value problem $y'' + y = 4x + 10 \sin x$ $y(\pi)=0, y'(\pi)=2$